Sai Saketh Nagavalli

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SUMMARY

GenAI Data Scientist with **3+ years of experience** across product analytics, machine learning, NLP, and production AI. Experienced with **Python, SQL, PySpark, RAG, LLM evaluation, XGBoost, TensorFlow, AWS, and Docker**, working across **10M+ records** and large-scale member data. Strong background in experimentation, predictive modeling, retrieval systems, and production ML workflows.

SKILLS

Programming & Data: Python, SQL, PySpark, Pandas, NumPy, Apache Spark, Data Processing, Feature Engineering, Statistical Analysis, Hypothesis Testing
Machine Learning & Deep Learning: Scikit-learn, XGBoost, TensorFlow, Keras, Classification, Regression, Clustering, Neural Networks, Predictive Modeling, Model Evaluation, A/B Testing
Generative AI & NLP: LLMs, RAG, Embeddings, Vector Search, Reranking, Prompt Engineering, LLM Evaluation, Groundedness, Hallucination Evaluation, NLP, Sentiment Analysis, Text Classification, TF-IDF, HITL
MLOps & Cloud: AWS, Docker, ML Pipelines, Batch Inference, Model Deployment, Model Monitoring, Drift Detection, Feature Quality, Retraining
Product Analytics: Funnel Analysis, Cohort Analysis, Customer Segmentation, Experimentation, Behavioral Analytics, Recommendation Models

EXPERIENCE

Included Health | *GenAI Data Scientist* Jan 2026 – Present

- Architected a **GenAI member-navigation workflow** using **Python, RAG, and LLMs** for benefits and care questions, improving grounded response accuracy from **84% to 93%**.
- Developed **hybrid retrieval pipelines** across **1M+ provider, benefits, and clinical records** using vector search, metadata filtering, and reranking for more relevant member responses.
- Created an **LLM evaluation framework with 1,500+ test scenarios** covering groundedness, hallucination, safety, and escalation before production releases.
- Designed **provider-matching and care-recommendation models** using claims, member preferences, and provider-quality data, improving recommendation precision by **17%**.
- Analyzed **5M+ member interactions** with **Python, SQL, and PySpark** to identify engagement patterns and guide personalization and care-navigation experiments.
- Partnered with **Product, ML Engineering, Analytics, and clinical teams** to productionize GenAI workflows with **HITL, A/B testing, and monitoring**, reducing roughly **400+ manual reviews per month**.

Neon IT Systems | *Product Data Scientist* Feb 2022 – Jul 2024

- Processed **10M+ user, transaction, and product-interaction records** using **Python, SQL, Pandas, and Spark**, producing reusable datasets and features for analytics and machine learning.
- Trained **classification, regression, and XGBoost models** for churn, conversion, and user propensity, supporting targeting and retention decisions across multiple product segments.
- Segmented customers through behavioral cohorts, funnels, clustering, and lifecycle metrics, helping Product teams identify high-value users and prioritize feature improvements.
- Evaluated onboarding, engagement, and recommendation experiments through **A/B testing**, defining success metrics and translating statistically significant findings into product decisions.
- Engineered **NLP workflows** for support tickets, surveys, and user feedback using **TF-IDF, embeddings, sentiment analysis, and text classification**, processing **500K+ text records**.
- Modeled behavioral and text-prediction use cases with **TensorFlow/Keras neural networks**, benchmarking performance against statistical and tree-based baselines.
- Automated feature processing, model validation, training, and batch inference through reusable **ML pipelines**, reducing manual effort across recurring modeling workflows.
- Productionized trained models as **Docker-based Python services on AWS**, supporting repeatable releases and integration with downstream applications.
- Mentored junior analysts while coordinating with **Product, Engineering, and Data teams** on model performance, drift, feature quality, prediction distributions, retraining, and data-quality fixes.

EDUCATION

Florida Atlantic University | Boca Raton, FL *Aug 2024 – May 2026*
Master of Science in Data Science and Analytics

CERTIFICATIONS

AWS Academy Data Analytics • AWS Academy Cloud Foundations • PCAP – Programming Essentials in Python (Cisco) • Data Analysis with Python (freeCodeCamp) • Data Analytics Virtual Internship (EduSkills Foundation)